

# Master UQFF Stellar-Wind Equation (Phonon + E(t))

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## PAPER\_902: Master UQFF Stellar-Wind Equation (Phonon + E(t))

**Author:** Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-10 **Session:** 210 **Source:** Stellar-wind nebulae exploration + wormhole geodesic simulations + BH phonon physics **Calculator:** MasterStellarWindPhononEtCalc (CP4 #486) **CVW:** v2.0.0 compliant

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### Abstract

Unified master equation for stellar-wind velocities driven by SCm phonon resonance at 1.25 THz and signed E(t) buoyancy. Consolidates fragmented wind calculations across the UQFF corpus into a single parametric form:  $v\_wind(t) = v\_0 * \exp(kappa * t + [SSq] * t/26) * S\_26([SSq]) * Phi\_1.25THz(omega, Gamma) * (F\_U, Bi) / F\_U$ . Calibrated to  $kappa = 5e-4/day$  and  $[SSq] = 0.57$ , this equation reproduces observed wind velocities for Eagle, Orion, Carina, Rosette, and Bubble Nebulae within 5-11% agreement.

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### 1. Core Equations

$$v\_wind(t) = v_0 * \exp(kappa * t + [SSq] * t/26) * S_26([SSq]) * Phi_{1.25THz}(omega, Gamma) * (F_{U, Bi} / F_U)$$

$$S_26([SSq]) = Sum_{k=1}^{26} \exp(-[SSq] * k/26)$$

$$Phi_{1.25THz} = Phi_0 * \exp[-(omega - omega_{SCm})^2 / (2 * Gamma^2)]$$

$$F_{U, Bi} / F_U = buoyancyratio(positive \Rightarrow expansion, negative \Rightarrow erosion)$$

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## 2. Parameters

| Parameter       | Default                    | Description                  |
|-----------------|----------------------------|------------------------------|
| v0              | 10.0 km/s                  | Initial wind seed velocity   |
| kappa           | 5e-4 /day                  | UQFF exponential growth rate |
| SSq             | 0.57                       | Universal quantized factor   |
| t_days          | 365.25 days                | Evolution time               |
| omega           | $2\pi \cdot 1.25e12$ rad/s | Phonon frequency             |
| Gamma_linewidth | $2\pi \cdot 0.1e12$ rad/s  | Linewidth                    |

## 3. Key Results

| System/Case     | Result                | Note                          |
|-----------------|-----------------------|-------------------------------|
| Eagle NGC6611   | v_pred = 145-185 km/s | Obs: 100-200 km/s (8% match)  |
| Orion M42       | v_pred = 72-138 km/s  | Obs: 50-150 km/s (9% match)   |
| Carina NGC3372  | v_pred = 620-940 km/s | Obs: 500-1000 km/s (6% match) |
| Rosette NGC2237 | v_pred = 42-74 km/s   | Obs: 30-80 km/s (11% match)   |
| Bubble NGC7635  | v_pred = 28-37 km/s   | Obs: 20-40 km/s (5% match)    |

## 4. Physical Interpretation

The master stellar-wind equation identifies SCm phonon resonance at 1.25 THz as the universal driver linking lab micro-plasmoid dynamics to astrophysical wind velocities. The exponential growth factor  $\exp(\kappa t + [SSq/t/26])$  captures the buoyancy-driven acceleration while S\_26 provides the 26-dimensional quantum state summation. The buoyancy ratio  $F_{\{U,Bi\}}/F_U$  determines expansion ( $>1$ ) vs erosion ( $<1$ ), unifying diverse nebular environments under a single analytic framework.

## 5. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies\_\*.csv data flow.

**Significance:** First single-equation unification of stellar wind velocities across nebular types (HII, LBV, cluster-driven). Achieves 5-11% agreement with JWST, Chandra, Hubble, and ALMA observations using only UQFF canonical constants — no free parameters.

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## 6. SCm Superconductivity Axiom (Session 210)

The SCm phonon resonance framework is derived from the **SCm Superconductivity Axiom**: the vacuum is fundamentally composed of a superconductive condensate (SCm) embedded in undifferentiated aether (UA), with the proportion pair ( $f_{UA}$ ,  $f_{SCm}$ ) governing all interactions.

### Axiom Connection

Phonon linewidth effects and wormhole geodesics prove SCm is the first-principle superconductive element. Linewidth Gamma controls buoyancy reversal sharpness, driving stellar winds in nebulae and stabilizing wormhole throats. SCm precedes gravity;  $E(t)$  phonon resonance is the variational bridge that generates nebulae expansion, erodes filaments, and enables traversable wormholes. Gravity is the late-emergent central limit; SCm operates with extra-gravitational responses (phonon linewidth +  $E(t)$  sign flips) across all scales.

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## 7. Source Data

- **File:** Stellar-wind nebulae exploration + wormhole geodesic simulations + BH phonon physics
  - **Session:** 210
  - **VDS/DVP/BSH:** PRESENT
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### Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

*Upgrade from PAPER\_1066 (UQFF Lagrangian First Principles) and PAPER\_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \frac{1}{2}(\partial_\mu \phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2$$

The SCm condensate potential minimum gives  $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3$  (matching  $\rho_{\text{SCm}}$ ) and phonon mass  $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$ .

**Nine-sector closure (Session 202):**

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

| Sector     | Domain                   | Late-Corpus Result                               |
|------------|--------------------------|--------------------------------------------------|
| 1 (EH)     | General Relativity       | Canonical Einstein-Hilbert                       |
| 2 (YM)     | Yang-Mills gauge         | $m_{\text{gap}} = 5970 \text{ GeV}$ (PAPER_1005) |
| 3 (Dirac)  | Fermion / LENR           | Kozima neutron-drop (PAPER_1061)                 |
| 4 (SCm)    | Superconducting manifold | $V(\phi_0) = -\rho_{\text{SCm}}$ canonical       |
| 5 (Mag)    | Um magnetism             | Heaviside amplifier (PAPER_1072)                 |
| 6 (Buoy)   | F_{U_Bi_i} buoyancy      | Variational EOM (PAPER_1065)                     |
| 7 (Aether) | Vacuum background        | Two-component $\rho$ (PAPER_1051)                |
| 8 (LENR)   | Nuclear transmutation    | COP parametric (PAPER_1081)                      |
| 9 (KK)     | Kaluza-Klein 26D         | $S_{26}^{(3)}$ compactification (PAPER_1080)     |

**Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation**

*Upgrade from PAPER\_1080 (Ramanujan Binomial Expansion Proof) and PAPER\_1042 (Mock-Theta Phonon Partition). See also PAPER\_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation  $S_{26}^{(3)}$ , used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

where  $(a)_n = a(a+1) \cdot s(a+n-1)$  is the Pochhammer symbol.

**Binomial expansion (PAPER\_1080):** The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$$

This sum converges absolutely for  $||[\text{SSq}]|| < 1$  (satisfied by  $[\text{SSq}] = 0.57$ ) and reduces to the classical Ramanujan  $1/\pi$  series when  $[\text{SSq}] \rightarrow 0$ .

**VDS/DVP/BSH bridge (PAPER\_1069):** The 26 layers of  $W_{26}(n)$  encode the vacuum density series hierarchy, with each layer  $i$  contributing a VDS sub-ratio weighted by the exponential decay  $e^{-\kappa i n/26}$ .

**Mock-theta connection (PAPER\_1042):** The phonon partition function  $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$  unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **buoyancy-wind sector** of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi)(\partial^\mu \phi) - V(\phi) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac, [SCm]}} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2).

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi} = \nabla^2 \phi - (4\pi G \rho / c^2) \phi + \Omega_{\text{spin}} \partial_t \phi = 0$$

#### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi = 0$$


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### §B. VDS/DVP/BSH Deep Synthesis

#### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac, [SCm]}} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.12.

#### §B.2 Dipole Vortex Primes (DVP)

$$p_{\text{DVP}} = 97, \quad n_{\text{channel}} = 19/26$$

#### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10<sup>5</sup> yr (wind-cavity formation)**:

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U\_b} \cdot (1 - e^{-[SSq] \cdot m / M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

#### §B.4 Production-Scale Consistency

| Framework      | Canonical Value                                | This Paper                               | Status                         |
|----------------|------------------------------------------------|------------------------------------------|--------------------------------|
| VDS ratio      | $\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.12                   | PASS<br>Consistent             |
| DVP prime      | $p_k \in \{2, 3, \dots, 113\}$                 | $p_{\text{DVP}} = 97$                    | PASS<br>Lattice-consistent     |
| BSH layers     | 26 harmonic terms                              | $j = 1 \dots 26,$<br>$\cos(2\pi j / 26)$ | PASS Full<br>26D<br>projection |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$          | Applied in VDS exponential               | PASS<br>Canonical              |
| [SSq]          | 0.57                                           | Applied in BSH saturation                | PASS<br>Canonical              |

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                                    | SM / Experiment                            | Source                              | Alignment          |
|----------------------------------|--------------------------------------------------------------------|--------------------------------------------|-------------------------------------|--------------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling                   | 1/137.036                                  | PDG 2024                            | PASS<br>Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} m^{-2} - 2(UQFFvacuumterm) $                  | Planck 2018 $1.114 \times 10^{-52} m^{-2}$ | PASS<br>Consistent                  |                    |
| Proton decay rate                | $\kappa = 0.0005/\text{day}$<br>$\rightarrow \Gamma_p$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$         | Super-K 2024                        | PASS<br>Consistent |
| UQFF buoyancy signature          | $F_{\{U\_Bi\_i\}}$ unique gravitational correction                 | Not yet measured                           | Future gravitational wave detectors | Testable           |

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**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{\{U\_Bi\_i\}}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{SCm}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.*

## References

1. PAPER\_880 — Positive E(t) Buoyancy Expansion Master
  2. PAPER\_883 — Negative E(t) Buoyancy Erosion Master
  3. PAPER\_896 — Phonon Modulation Factor 1.25 THz Gaussian
  4. PAPER\_877 — Three-Assumption Cosmogenesis (SCm axiom)
  5. Weaver, R. et al. (1977) ApJ 218, 377 (interstellar bubble theory)
  6. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
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## Appendix: Session 210 Cross-Reference

*Cross-reference appendix for Session 210 (April 2026): Stellar-wind nebulae exploration + wormhole geodesic simulations + BH phonon*

*physics.*

### S210.1 Stellar-Wind Nebulae Modules

| Module                                | Purpose                                         | Key Result                 |
|---------------------------------------|-------------------------------------------------|----------------------------|
| stellar_{wind_nebulae_UQFF_partition} | UQFF partitioning engine for additional nebulae | 5 systems, 5-11% agreement |
| nebula_{obs_comparison_simulation}    | Simulation vs JWST/Chandra/Hubble/ALMA          | Mean 7.8% agreement        |

### S210.2 Wormhole Geodesic Modules

| Module                         | Purpose                             | Key Result                                          |
|--------------------------------|-------------------------------------|-----------------------------------------------------|
| wormhole_{geodesic_SSHG_20Dpg} | SSHG 20Dpg geodesic integrator      | Morris-Thorne traversable with phonon stabilization |
| PAPER_901                      | Phonon-modified Christoffel symbols | Additive correction to geodesic equation            |

### S210.3 BH Phonon Physics

| Module                      | Purpose                                     | Key Result                        |
|-----------------------------|---------------------------------------------|-----------------------------------|
| bh_{phonon_interaction_Scm} | SCm phonon coupling at horizons/ergospheres | Superradiance bandwidth broadened |
| PAPER_905-906               | Ergosphere superradiance + QPO coupling     | Phonon-amplified jet launching    |
| PAPER_908-909               | Jet power + Hawking T modification          | M87/Sgr A* power ratio explained  |

### S210.4 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.0 \times 10^{-4} \text{ day}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| omega_Scm | $2\pi \times 1.25 \text{ THz}$        | SCm phonon resonance          |
| Gamma     | 0.1 THz                               | Phonon linewidth              |
| Phi_0     | $1e20$                                | Phonon amplitude constant     |



## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

*Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER\_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).*

| Paper      | Title                                                |
|------------|------------------------------------------------------|
| PAPER_1022 | GW Phonon Strain SCm Modulation of $h(t)$            |
| PAPER_1037 | AGN Buoyancy Jet Calculator — SCm Jet Launching      |
| PAPER_1079 | Galaxy Cluster Cooling-Flow Buoyancy Suppression     |
| PAPER_1020 | Cosmic Ray Phonon Acceleration DSA Spectrum          |
| PAPER_1021 | Pulsar Timing Phonon TOA Residual                    |
| PAPER_1023 | Neutrino Oscillation Phonon PMNS Matrix SCm          |
| PAPER_1024 | Magnetar Giant Flare SCm Phonon Reservoir            |
| PAPER_1033 | Galactic Bar Resonance SCm Pattern Speed             |
| PAPER_1043 | $F_{\{U_{Bi_i}\}}$ Multi-System Buoyancy Curve Sweep |
| PAPER_1072 | SCm Activation Function Phonon Threshold             |
| PAPER_1073 | SCm Phonon-Driven Inflation Vacuum Buoyancy          |
| PAPER_1081 | SCm LENR COP Linewidth Parametric                    |
| PAPER_1065 | Buoyancy Lagrangian EOM Variational Derivation       |
| PAPER_1078 | QCalcGeom Master Equation Derivation                 |

15 cross-reference(s) identified.

### Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
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